SIMATS ENGINEERING
CO1 - ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA0405

Name of the Student	**Narra Sai Mounika**
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GitHub Link	**https://github.com/saimounika1315/MLA0403---Deep-Learning**

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

CO1 Weightage: 15%
Assessment Tool Weightage: 35%
Duration: 1 Hour
Marks: 50

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. **A coin is tossed 10 times and the outcomes are:**
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
   Answer:
   Given outcomes:
   H H T H H H T H H T
   Number of Heads (H) = 7
   Total Tosses (n) = 10
   MLE for probability of Heads:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

$$\hat{p} = \frac{7}{10} = 0.7$$

   MLE of probability of Heads = 0.7 (70%)

2. Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Answer:

Given:

Passed Students = 42

Total Students = 50

MLE for probability of passing:

$$\hat{p} = \frac{42}{50}$$
$$\hat{p} = 0.84$$

MLE of passing probability = 0.84 (84%)

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.

Find the MLE for the probability that a bulb is defective.

Answer:

Given:

Defective Bulbs = 8

Total Bulbs = 100

MLE for defective probability:

$$\hat{p} = \frac{8}{100}$$
$$\hat{p} = 0.08$$

MLE of defective bulb probability = 0.08 (8%)

4. A biased die is rolled 60 times. The number 6 appears 18 times.

Estimate the probability of rolling a 6 using MLE.

Answer:

Given:

Number of times 6 appears = 18

Total Rolls = 60

MLE for probability of rolling 6:

$$\hat{p} = \frac{18}{60}$$
$$\hat{p} = 0.3$$

MLE of rolling a 6 = 0.3 (30%)

5. Given:

Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
y = w × x

1. **Find the predicted output.**
2. **Calculate the error.**
3. **If the gradient is 4 and learning rate is 0.01, find the updated weight.**

Answer:

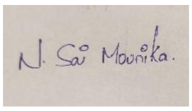
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Gradient = 4
Learning Rate = 0.01
Step 1: Predicted Output

$$y = w \times x$$
$$y = 2 \times 4 = 8$$

Predicted Output = 8

Code of Conduct:

I Narra Sai Mounika Reg no: 192425367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Narra Sai Mounika
Name of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation